Network design practice 02

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1 Overview

This practice simulates P4 code and network structure.

2 Assignment 1

```
struct metadata {
    /* empty */
}

struct headers {
    ethernet_t   ethernet;
    ipv4_t      ipv4;
                        /* Assignment1 TODO: add IPv4 header */
}
```

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Figure 1: Adding IPv4 header

Added IPv4 header in structure headers referencing how ethernet header is added.

```
    state parse_ipv4 {
        packet.extract(hdr.ipv4);              /* Assignment1 TODO: fill in argument to specify which header to extract */
        transition accept;
    }

}

/*   C H E C K S U M    V E R I F I C A T I O N   */

control MyVerifyChecksum(inout headers hdr, inout metadata meta) {
    apply {  }
}

/*  I N G R E S S   P R O C E S S I N G */

control MyIngress(inout headers hdr,
                  inout metadata meta,
                  inout standard_metadata_t standard_metadata) {
    action drop() {
        mark_to_drop(standard_metadata);
    }

    action ipv4_forward(macAddr_t dstAddr, egressSpec_t port) {
        standard_metadata.egress_spec = port;                /* Assignment1 TODO: set egress port */
        hdr.ethernet.srcAddr = hdr.ethernet.dstAddr;                 /* Assignment1 TODO: set source MAC address */
        hdr.ethernet.dstAddr = dstAddr;                              /* Assignment1 TODO: set destination MAC address */
        hdr.ipv4.ttl = hdr.ipv4.ttl - 1 ;                                    /* Assignment1 TODO: decrement ttl by 1 */
    }

    table ipv4_lpm {
```

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Figure 2: Adding IPv4 header

I passed the argument hdr.ipv4 for packet to extract. See ethernet header is extracted in similar manner. On the lower block, we define forwarding action. The egress port is set to port (the one passed from the argument), the source MAC address would be the one previously the destination address, since the packet was freshly forwarded to the switch when the action is triggered. The destination address is set to dstAddr which is also from the argument. Lastly, ttl was decremented by 1.

```
  tail -f /home/p4/tutorials/exercises/basic/logs/<switchname>.log

To view the switch output pcap, check the pcap files in /home/p4/tutorials/exercises/basic/pcaps:
 for example run:  sudo tcpdump -xxx -r s1-eth1.pcap

To view the P4Runtime requests sent to the switch, check the
corresponding txt file in /home/p4/tutorials/exercises/basic/logs:
 for example run:  cat /home/p4/tutorials/exercises/basic/logs/s1-p4runtime-requests.txt

E0326 12:02:00.632631913   27767 fork_posix.cc:63]           Fork support is only compatible with the
mininet> ping all
*** Unknown command: ping all
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet> S
```

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We observe all packets are sent and received.

2.1 Assignment 2

```
      "action_name": "MyIngress.ipv4_forward",
      "action_params": {
        "dstAddr": "08:00:00:00:02:22",
        "port": 2
      }
    },
    {
      "table": "MyIngress.ipv4_lpm",
      "match": {
        "hdr.ipv4.dstAddr": ["10.0.3.3", 32]
      },
      "action_name": "MyIngress.ipv4_forward",
      "action_params": {
        "dstAddr": "08:00:00:00:03:00",
        "port": 4
      }
    },
    {
      "table": "MyIngress.ipv4_lpm",
      "match": {
        "hdr.ipv4.dstAddr": ["10.0.4.4", 32]
```

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Figure 3: Modifying port

Since the link between s3 and s1 is down, I decided to redirect all connection that involves s3 to s4. Naturally, if the packet destination is either h3 or h4, s1 forwards packet to port 4, not 3.

```
    {
      "table": "MyIngress.ipv4_lpm",
      "match": {
        "hdr.ipv4.dstAddr": ["10.0.1.1", 32]
      },
      "action_name": "MyIngress.ipv4_forward",
      "action_params": {
        "dstAddr": "08:00:00:00:03:00",
        "port": 3
      }
    },
    {
      "table": "MyIngress.ipv4_lpm",
      "match": {
        "hdr.ipv4.dstAddr": ["10.0.2.2", 32]
      },
      "action_name": "MyIngress.ipv4_forward",
      "action_params": {
        "dstAddr": "08:00:00:00:04:00",
        "port": 3
      }
    },
    {
```

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Figure 4: Modifying port

Similarly, all packets bound for h1 and h2, s2 forwards packet to port 3.

```
E0326 17:21:58.499416764    3005 fork_posix.cc:63]
mininet> link s1 s3 down
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet> S
```

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We observe all packets are sent and received.